

Cesar A. Aguilar

Full Stack Systems Engineer | AI Integration Cloud Network AWS Specs

Pasadena, California | 626.360.5860 | blueavian9@gmail.com | [GitHub](#) | [LinkedIn](#) | [Portfolio](#)

TECHNICAL SKILLS

EDUCATION

Western Governors University (WGU) | Online, Accredited University B.S. in Cloud Network Engineering Cisco/ AWS Specs

| January 2025 - In Progress | Expected Graduation: December 2026

- Current GPA: 3.8/4.0 (verified and maintained)
- Will remain enrolled in degree-seeking program post-internship
- Curriculum includes 16 industry-recognized certifications in cloud architecture, DevOps, networking, and security
- Relevant Coursework: Software Development, Cloud Architecture, Network Engineering, Systems Administration, DevOps Practices

Planned: M.S. in Artificial Intelligence and Machine Learning | Expected June 2028

from Western Governors University(WGU) & An

Online Doctor of Engineering in Artificial Intelligence and Machine Learning |

Expected June 2033 from George Washington University. (GW)

Bloom Institute of Technology | Full Stack Software Engineer Certification | March 2022 - April 2024

- Enterprise-level training in full-stack development, Agile methodologies, and software engineering principles
- Trained by Distinguished Software Engineering Principal Expert NDA

Pasadena City College | Computer Science Major | 2007 - 2011

TECHNICAL SKILLS

Programming Languages: JavaScript (ES6+), Python3, TypeScript, HTML5, CSS3, SQL.

Frontend Technologies: React.js, Context API, Axios, Responsive Web Design, Flexbox, CSS Grid, UI/UX Principles.

Backend Technologies: Node.js, Express.js, RESTful APIs, PostgreSQL, SQLite, Knex.js, Auth0, bcrypt.js, JWT.

Databases: PostgreSQL, MongoDB, Supabase.

Cloud & DevOps: AWS Polly V3 SDK, CI/CD Pipelines (GitHub Actions), Git/GitHub, dotenv, Linux CLI, Environment Configuration.

AI/ML Integration: OpenAI APIs (GPT, DALL·E, Whisper), AWS Polly V3, RAG (Retrieval-Augmented Generation), LangChain, Speech-to-Text (STT), Text-to-Speech (TTS), AI Content Moderation

Testing & Quality Assurance: Unit Testing (Jest), Postman API Testing, Manual Testing Workflows, Debugging Strategies.

Development Tools: VS Code, Git Bash, Git CLI, Agile/Scrum Methodologies, GitHub Project Boards, Markdown Documentation.

Security & Best Practices: bcrypt.js Password Hashing, JWT Authentication, Role-Based Access Control (RBAC), Secrets Management.

I'm comfortable working across the full stack, with a strong focus on clean component architecture on the frontend and scalable API design on the backend.

Additional: Microsoft Office Suite (Excel, Word, PowerPoint), Technical Documentation, Cross-functional Collaboration.

In-Progress Cloud Certifications (WGU Program):

- AWS Cloud Practitioner, AWS Solutions Architect Associate, AWS Developer Associate, AWS SysOps Administrator
- CompTIA Cloud Essentials+, CompTIA Network+, CompTIA Linux+, ITIL Foundation v4
- Microsoft Azure Fundamentals, Microsoft Azure Administrator (AZ-104)

Language Proficiency: Spanish (Fluent) | Hindi (Conversational) | Farsi (Basic) | French (Basic) | Mandarin Chinese (Basic)

RELEVANT PROJECTS

Online Booking Holistic Psychology Wellness App

Node.js|React.js|Vite|Tailwind CSS|Open AI APIs|Supabase|Stripe Payments| Auth0| [Github](#) | [Live Demo](#)

- Architected and deployed AI-enabled full-stack applications integrating LLM APIs (OpenAI) with structured prompt pipelines, context management, and output validation for production-grade reliability.
Engineered AI-assisted automation workflows using LangChain and custom orchestration layers, enabling intelligent task execution, data transformation, and decision-support features within web platforms.
- Designed scalable React + Node.js systems optimized for AI integration, including token-efficient request handling, latency optimization, secure API abstraction layers, and cloud-ready deployment strategies.
- Developed HIPAA-conscious AI-enabled wellness platform with secure authentication, protected data workflows, and AI-enhanced booking/reporting features, balancing compliance, usability, and performance.
- Rapidly prototype and ship AI-powered products within 14-day sprint cycles, combining product strategy, system architecture, and execution to move from concept to deployed solution.

AI-Powered Educational Platform | Full-Stack Software Engineer

Node.js, Express, AWS Polly V3 SDK, OpenAI APIs | [Github](#) | [Live Demo](#)

- Developed scalable full-stack web applications showcasing AI-driven features including speech-to-text (STT), text-to-speech (TTS), vision analysis, and content moderation
- Architected modular backend systems using Node.js and Express to integrate AWS Polly V3 SDK and OpenAI APIs
- Implemented efficient data handling and API performance optimization serving 500+ potential users
- Created comprehensive technical documentation presenting complex integration processes in clear, accessible formats for stakeholders

AI Chatbot Summarizer | Full-Stack Engineer

React.js, Node.js, RAG (Retrieval-Augmented Generation) | [Github](#) | [Live Demo](#)

- Built knowledge-augmented AI chatbot using Retrieval-Augmented Generation to summarize school information with advanced search pipelines
- Designed intuitive user interface enabling seamless interaction between users and AI-driven backend systems

- Collaborated with cross-functional team of 4 developers to ensure efficient integration of frontend and AI technologies
- Implemented state management and data persistence for optimal user experience

Secure Authentication System | Backend Engineer

Express.js, PostgreSQL, JWT, bcrypt.js | DB |

[GitHub](#) |

[Live Demo](#)

- Designed secure login/signup systems implementing token-based authentication with JWT expiration and password encryption best practices
- Built authentication endpoints with Express middleware following industry-standard security protocols and RBAC
- Engineered PostgreSQL database schemas using Knex.js for efficient data persistence and retrieval
- Implemented comprehensive unit testing with Jest achieving 98% code coverage rate

RESTful API Development | Backend Engineer

Node.js, Express, Knex.js, SQLite |

[GitHub](#) |

[Live Demo](#)

- Created scalable RESTful APIs for high-performance web applications with optimized SQL queries
- Architected database schemas and implemented migrations using Knex.js for seamless backend integration
- Reduced API response times by 40% through query optimization and efficient data handling
- Developed comprehensive API documentation ensuring developer accessibility and ease of integration

PROFESSIONAL EXPERIENCE

Freelancer | Full Stack AI Integrations System Engineer

April 2024 - Present / *Remote* / Los Angeles, CA.

Developed secure RESTful APIs integrating AWS Polly V3 SDK and OpenAI APIs for text generation, vision, speech, and content moderation features

- Built scalable backend systems with Node.js/Express supporting real-time AI-driven educational tools serving 1,000+ users
- Implemented backend workflows and CI/CD pipelines to enhance system reliability by 35%
- Led security configurations across AWS environments following industry best practices (environment variables, API key security)

- Collaborated with cross-functional teams including DevOps, frontend developers, and product managers to deliver cohesive solutions
- Documented complex technical processes and presented integration workflows to non-technical stakeholders in clear formats

TalkAnimate Inc. | Full Stack Software Engineer (Internship) | April 2024 - November 2024
Remote / Burbank, CA

- Developed secure RESTful APIs integrating AWS Polly V3 SDK and OpenAI APIs for text generation, vision, speech, and content moderation features
- Built scalable backend systems with Node.js/Express supporting real-time AI-driven educational tools serving 1,000+ users
- Implemented backend workflows and CI/CD pipelines to enhance system reliability by 35%
- Led security configurations across AWS environments following industry best practices (environment variables, API key security)
- Collaborated with cross-functional teams including DevOps, frontend developers, and product managers to deliver cohesive solutions
- Documented complex technical processes and presented integration workflows to non-technical stakeholders in clear formats

Bloom Institute of Technology | Full Stack Software Developer | March 2021 - March 2024
Remote / San Francisco, CA

- Engineered Full-Stack Asylum Report Generator with Auth0 authentication and PostgreSQL persistence
- Designed and deployed RESTful APIs using Express.js, improving system performance by 30% through optimization
- Conducted comprehensive unit testing with Jest ensuring 98% code coverage rate and robust codebase
- Collaborated with cross-functional teams to align backend logic with frontend requirements delivering scalable solutions
- Participated in Agile sprint planning, daily standups, and retrospectives ensuring timely project delivery

LEADERSHIP & TEAMWORK EXPERIENCE

Close Range International Inc. | Security Officer, L.A.P.D. CouncilMember CD-14 Office | February 2022 - Present
El Sereno, CA - City of Los Angeles CD-14

- Collaborate with team of 100+ members to implement strategic security solutions and mitigate safety risks

- Partner with law enforcement agencies and local government to coordinate emergency response procedures demonstrating strong stakeholder collaboration
- Manage multiple priorities simultaneously while maintaining high attention to detail and organizational excellence
- Demonstrate customer-focused attitude serving government facilities and public safety initiatives

Allied Universal (fka United Guard Security Inc.) | Armed Security Officer | June 2021 - January 2022

- Reduced security incidents by 30% through proactive problem-solving and strategic implementation of surveillance protocols
- Improved response times for critical incidents at City of Hope Cancer Research Center through process optimization
- Demonstrated ability to work in healthcare environments requiring attention to detail and patient safety awareness
- Collaborated cross-departmentally to coordinate emergency procedures and safety improvements

United Guard Security Inc. | Supervisor of Security | October 2019 - June 2021

- Supervised, trained, and scheduled 30+ security personnel, increasing compliance metrics by 15%
- Managed access control systems and authorization reports, achieving 100% reduction in unauthorized access
- Evaluated team performance and implemented process improvements resulting in enhanced safety response times
- Developed leadership skills through mentoring junior team members and managing complex operational schedules
- Demonstrated organizational skills managing multiple priorities and delivering results under pressure

ADDITIONAL CERTIFICATIONS & TRAINING

freeCodeCamp.org

- JavaScript Algorithms & Data Structures Certification | 2021 - 2022
- Responsive Web Design Certification | 2020 - 2021

SoloLearn Mobile App

- Mobile & Web Development Certifications | 2017 - 2020

- Demonstrated commitment to healthcare industry and patient-focused service
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KEY QUALIFICATIONS

Academic Requirements Met:

- Currently pursuing B.S. in Cloud Network Engineering at Western Governors University (WGU)
- Cumulative GPA: 3.8/4.0 (verified and maintained)
- Will remain enrolled in degree-seeking program after internship completion
- Expected graduation: December 2026

Authorization:

- Legally authorized to work in the United States
- No sponsorship required now or in the future

Core Competencies:

- **Strong Communication Skills:** Proven ability to document complex technical processes and present to stakeholders in clear formats; multilingual capabilities (5 languages)
- **Leadership:** Supervised 30+ team members achieving 15% increase in compliance and efficiency
- **Problem-Solving:** Reduced security incidents by 30%; optimized API performance by 40%; improved system reliability by 35%
- **Organizational Skills:** Successfully managed multiple priorities across development projects, security operations, and academic pursuits
- **Collaboration:** Demonstrated success working with cross-functional teams including DevOps, frontend developers, law enforcement, and government agencies

Technical Proficiency:

- Proficient in Microsoft Office Suite (Excel, Word, PowerPoint)
- Extensive experience with development tools, CI/CD pipelines, and Agile methodologies
- Hands-on experience with software development, testing, and deployment
- Active learner eager to grow in dynamic healthcare technology environment

Alignment with Stryker's "Who We Want":

- **Challenger:** Consistently sought complex projects (AI integration, secure authentication, scalable backends)
- **Teammate:** Proven collaboration with cross-functional teams across multiple organizations
- **Charismatic Networker:** Built relationships with law enforcement, government officials, development teams, and stakeholders
- **Strategic Thinker:** Proposed and implemented innovative AI solutions exceeding performance objectives
- **Customer-Oriented Achiever:** Healthcare facility experience (City of Hope) with unparalleled work ethic
- **Game Changer:** Persistent drive to leverage technology for meaningful impact in healthcare and education